



بسم الله الرحمن الرحيم
جامعة كركري - كلية علوم الحاسوب وتقانة المعلومات
المستوى الثالث (هـ. برمجيات)
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الزمن: ثلاث ساعات

المادة: فحص البرمجيات وضمان الجودة
أستاذ المادة: د. كمال بشير

Answer all questions.

Q¹. True/False Indicate T (True) or F (False):

1. Software Quality Assurance is solely the responsibility of the QA team. ()
2. Cyclomatic Complexity measures the number of independent paths in a program. ()
3. Function Points (FP) are language-dependent. ()
4. Validation ensures the software meets user requirements. ()
5. Static testing involves executing the code. ()
6. A Test Plan describes how to perform specific tests. ()
7. Defect Removal Efficiency (DRE) = Defects / (Defects + Errors). ()
8. White Box Testing focuses on internal code structure. ()
9. MTTR stands for Mean Time To Repair. ()
10. Automated testing is ideal for usability testing. ()
11. Regression testing ensures new changes do not break existing functionality. ()
12. A "bug" and a "defect" mean the same thing. ()
13. Configuration Management prevents version control issues. ()
14. Test cases should only include valid input conditions. ()
15. McCabe's Complexity Metric uses control flow graphs. ()
16. Quality Metrics are always qualitative. ()
17. Verification includes activities like code reviews. ()
18. Risk-Based Testing prioritizes testing for high-risk features. ()
19. Manual testing is more efficient for performance testing. ()
20. A Test Case includes expected results. ()

Q*. Definitions

1. Software Quality
2. SQA (Software Quality Assurance)
3. Defect Removal Efficiency (DRE)
4. Cyclomatic Complexity
5. Function Point (FP)
6. Verification
7. Validation
8. Black Box Testing
9. White Box Testing
10. Static Testing
11. Dynamic Testing
12. Test Plan
13. Test Case
14. Bug Life Cycle
15. MTBF (Mean Time Between Failures)
16. McCabe's Complexity Metric
17. Configuration Management
18. Regression Testing
19. Quality Metrics
20. Risk-Based Testing

Q 3. Direct Answers: Answer briefly (1-2 sentences):

1. What is the purpose of a Test Plan?
2. Differentiate between Verification and Validation.
3. What is Cyclomatic Complexity used for?
4. List two advantages of Function-Oriented Metrics.
5. Explain the term "Defect Removal Efficiency (DRE)."
6. What is the difference between static and dynamic testing?
7. Name two phases of the Bug Life Cycle.
8. Why is Configuration Management important?

9. What are the three types of software maintenance?
10. What is the main challenge of Test Automation?

Q 4. Discussion/Comparison/Drawing

1. Compare Black Box Testing and White Box Testing.
2. Discuss the importance of Software Quality Assurance in Agile projects.
3. Draw the Bug Life Cycle and label its phases.
4. Explain how Cyclomatic Complexity impacts code maintainability.
5. Compare Manual Testing and Automated Testing in terms of cost and efficiency.
6. Describe the steps to calculate Function Points (FP).
7. Analyze the role of metrics in improving software quality.
8. Discuss the challenges of implementing Test Automation.
9. Explain the concept of "Risk-Based Testing" with an example.
10. Compare Verification and Validation in the context of software testing.

Good Luck!